

Braid Presentations for Concurrent Programs

Zachary Stone

September 22, 2026

Abstract

Ziemiański’s cube chain category $\text{Ch}(K)$ is a combinatorial model for the space of directed paths on a precubical set K . Paliga and Ziemiański built a braid invariant for precubical sets by observing a relation between $\pi_1|\text{Ch}(Z)|$ and braids, where Z is the terminal precubical set. However, that invariant trivializes for Euclidean K because $|\text{Ch}(I^n)|$ is contractible [PZ21, Prop. 6.1]. This paper constructs a localization of $\text{Ch}(K)$ at morphisms that preserve the order of coordinates, called $\text{Br}(K)$. And we will show that this structure is the source of the braid structure of $\pi_1|\text{Ch}(Z)|$. But, there is no nerve or geometric realization in this technique. Instead we build a 2-polygraph giving an artin braid-like presentation of $\text{Br}(K)$. The $\{0, 1, 2\}$ -cells of the polygraph are the dimension $\{0, 1, 2\}$ chains of K . Intuitively, 3-cells of K contribute braid relations, and pairs of non-intersecting 2-cells of K contribute commuting relations. This is not just a metaphor: at $K = Z$, we get the artin relations and generators exactly. So $\text{Br}(Z)$ is equivalent to the positive braid category. The presentation is natural in K , providing a generalization of the existing work. As an application, we establish a link to the hyperplane arrangement literature. The way braids are constructed via Salvetti’s theorem for the braid arrangement exactly matches the polygraph we build here for the cube. All this work is formalized in Lean [Sto26].

Contents

1	Introduction	2
1.1	Background	2
1.2	Results	3
1.3	Formalisation	4
1.4	Notation	4
2	Background	5
2.1	Cube chains	5
2.2	Wedge coordinates	6
2.3	Polygraphs and presentations	7
2.4	Localization	8
2.5	Newman’s lemma, coherent form	8
3	Squares, runs, and biclosed sets	9
3.1	Squares and triangles	9
3.2	Runs	9
3.3	Biclosed sets	10
4	Crossings, order-preserving refinements, and $\text{Br}(K)$	11

4.1	Crossings	11
4.2	Order-preserving refinements	11
4.3	The braiding	12
5	The runs over a chain	12
6	Steps, cells, and the paper polygraph	13
6.1	Steps	13
6.2	Covers	13
6.3	Factorisations and the cell a run rises into	14
6.4	The attachment of a 2-cell	15
6.5	The polygraph	16
7	Theorem A: the polygraph presents $\text{Br}(K)$	17
7.1	From the cells to the localization	17
7.2	From the chains to the cells	17
7.3	The two readings are inverse	18
8	Theorem B: naturality in K	19
9	Theorem C: at the point the polygraph is Artin's	19
9.1	The base	19
9.2	Where a run passes a step	20
9.3	The two words alternate	21
9.4	The comparison	21

1 Introduction

1.1 Background

Ziemiański's cube chain category $\text{Ch}(K)$ is a combinatorial model for the executions of a bipointed precubical set K [Zie20, Thm. 7.6]. Paliga and Ziemiański established a homotopy equivalence between $|\text{Ch}(Z)|$ and configuration spaces of points in the plane, where Z is the terminal precubical set. It extends, by naturality, to an invariant of any precubical set K . However, it trivializes on any Euclidean complex, see [PZ21, Prop. 6.1]. The proof is one line: the invariant factors through $|\text{Ch}(I^n)|$, which is contractible.

The braid structure that $\pi_1|\text{Ch}(Z)|$ sees is, however, already visible one level down, in $\text{Ch}(K)$ itself, and it does not need a nerve. An object of $\text{Ch}(K)$ is a decomposition of an execution into blocks of simultaneous events, and a morphism is a *refinement*: a relabelling of events which coarsens the blocking. Some refinements only glue adjacent blocks together and leave every event where it was; the rest genuinely reorder events across a block boundary. Call the first kind *order preserving* and write W for the class they form. Inverting them is the whole construction:

$$\text{Br}(K) := \text{Ch}(K)^{\text{op}}[(W^{\text{op}})^{-1}].$$

Nothing is quotiented. The order-preserving refinements are exactly those with no reordering to record, so inverting them turns $\text{Ch}(K)$ into a category whose arrows remember *which* events crossed

which, and how often. It is a braiding, not a permutation: two independent events that swap twice are not the identity.

The point of the paper is to write $\text{Br}(K)$ down by generators and relations, for every K , naturally in K , and to see that the answer at the point is Artin's presentation. The generators and relations are not invented: they are the low-dimensional chains of K .

1.2 Results

A chain of degree 0 is a *run*: an execution with a total order on its events. A chain e of any degree spans two runs, the run $\underline{e}$ it is glued from, which reaches it by an order-preserving refinement, and the run $\bar{e}$ that reverses every block of e . So every chain is an arrow-shaped piece of data, and we take the low-dimensional ones as cells.

Definition 1.1 (Definition 6.16). The **paper polygraph** $\mathcal{P}(K)$ is the 2-polygraph whose

- 0-cells are the runs of K ;
- 1-cells $X \rightarrow Y$ are the chains e of degree 1 with $\underline{e} = X$, $\bar{e} = Y$;
- 2-cells $X \rightarrow Y$ are the chains e of degree 2 with $\underline{e} = X$, $\bar{e} = Y$,

a 2-cell e being attached along the two maximal chains of the poset of runs over e (Proposition 6.14).

Theorem 1.2 (Theorem 7.9). *For every bipointed precubical set K , with no hypothesis on K , the polygraph $\mathcal{P}(K)$ presents $\text{Br}(K)$.*

Theorem 1.3 (Theorem 8.2). *$K \mapsto \mathcal{P}(K)$ is a functor $\square_* \rightarrow \mathbf{Poly}$, and the presentation of Theorem 1.2 is natural in K : for every $u : K \rightarrow L$ the square*

$$\begin{array}{ccc} \langle \mathcal{P}(K) \rangle & \xrightarrow{\langle \mathcal{P}(u) \rangle} & \langle \mathcal{P}(L) \rangle \\ E_K \downarrow & & \downarrow E_L \\ \text{Br}(K) & \xrightarrow{\text{Br}(u)} & \text{Br}(L) \end{array}$$

commutes on the nose, as an equality of functors.

Theorem 1.4 (Theorem 9.12). *At the terminal object Z the paper polygraph is Artin's, cell for cell and word for word: $\mathcal{P}(Z) \cong \mathcal{P}_{\text{Artin}}$, the polygraph with one 0-cell per strand count N , the $N - 1$ generators $\sigma_1, \dots, \sigma_{N-1}$, and one 2-cell per pair of generators carrying its two alternating words. Consequently $\text{Br}(Z) \simeq \mathbf{Braid}^{+, \text{op}}$, the positive braid category.*

Two things are worth saying about the shape of the answer. First, no braid theory goes into Theorem 1.4: the generators come out of $\text{Ch}(Z)$ as the degree-one chains, the relations as the degree-two chains, and the alternating words as the two ways of sweeping the squares of a degree-two chain. Artin's presentation [Art47] is the output. Second, the dichotomy in Artin's relations is the dichotomy between the two kinds of degree-two chain. A degree-two chain is either one block of dimension 3 or two blocks of dimension 2; the first spans a hexagon of runs and gives $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$, the second spans a square and gives $\sigma_i \sigma_k = \sigma_k \sigma_i$ (Proposition 6.15). So 3-cells of K contribute braid relations and disjoint pairs of 2-cells contribute commutations, as promised.

The nearest existing statement is Gaussent–Guiraud–Malbos’s coherent presentation of an Artin monoid [GGM15], which produces the same 2-cells from Garside theory. Here they are produced by the geometry of K , for every K , and the Artin case is the value at a point.

1.3 Formalisation

Everything below is formalised in Lean 4 on top of mathlib [Sto26]; the repository is sorry-free and the three theorems are checked with no hypothesis on K . The correspondence for the main statements:

statement	Lean declaration
$\text{Br}(K)$ (Definition 4.7)	<code>ChainCat.Br</code>
W (Definition 4.3)	<code>ChainCat.W</code>
$\mathcal{P}(K)$ (Definition 6.16)	<code>ChainCat.Paper.poly</code>
Theorem 7.9	<code>ChainCat.Paper.paperPresents</code>
Theorem 8.2	<code>ChainCat.Paper.polyFunctor, paperSquare</code>
Theorem 9.12	<code>ChainCat.Paper.paperArtinIso</code>
Corollary 9.13	<code>ChainCat.fullBaseEquiv</code>

The file `CubeChains/Paper.lean` is the paper read back into the tree: each definition and lemma below appears there as an `example` discharged by a single citation.

1.4 Notation

To fix notation, let **Cube** be the box category: its objects are the cubes I^n for $n \geq 0$, and its morphisms are the face embeddings (composites of the coface maps δ_i^ϵ). A *precubical set* is a presheaf on **Cube**; write $\square$ for the category of precubical sets. Concretely, a precubical set K has, for each $n \geq 0$, a set K_n of n -cells, together with face maps

$$d_i^\epsilon : K_n \rightarrow K_{n-1} \quad (1 \leq i \leq n, \epsilon \in \{0, 1\}),$$

where i names a direction and ϵ a side, subject to the precubical identities $d_i^\epsilon d_j^\eta = d_{j-1}^\eta d_i^\epsilon$ for $i < j$. We write I^n both for the standard n -cube as an object of **Cube** and for the presheaf it represents, and we set $[n] := \{1, \dots, n\}$, so that $[0] = \emptyset$. Yoneda identifies the two descriptions of a cell,

$$K_n = K(I^n) = \text{Hom}_\square(I^n, K),$$

and we pass between them without comment; in particular K_0 is the set of vertices.

We usually work with precubical sets carrying designated start and end vertices. A *bipointed* precubical set is a precubical set K with two chosen vertices $\text{init}_K, \text{fin}_K \in K_0$, and morphisms are required to preserve them; write $\square_*$ for the resulting category. Each cube I^n is bipointed by its initial vertex $(0, \dots, 0)$ and final vertex $(1, \dots, 1)$. We write Z for the final object of $\square_*$.

Convention 1.5. Named ambient categories are set in bold (**Cat**, **Set**, **Cube**, **Braid**⁺, **Poly**); constructions applied to an object are set upright ($\text{Ch}(K)$, $\text{Run}(K)$, $\text{Sq}(d)$). Functors are written with $\rightarrow$; we reserve $\Rightarrow$ for natural transformations. The standard n -cube is I^n ; the symbol $\square$ names the category of precubical sets. Calligraphic letters name constructions that are posets or polygraphs rather than categories: $\mathcal{R}(d)$, $\mathcal{P}(K)$. Composition is classical: $g \circ f$, and also gf , mean “first f , then g ”. A word in a polygraph is a path, read *left to right*, so the word $\sigma_i \sigma_k$ is the arrow $\sigma_k \circ \sigma_i$.

Remark 1.6 (Variance). Following the cube chain literature [PZ21; Zie20], a morphism of $\text{Ch}(K)$ runs from a *finer* chain to a *coarser* one, that is against the direction in which an execution is built up. So we localize the opposite category, $\text{Br}(K) := \text{Ch}(K)^{\text{op}}[(W^{\text{op}})^{-1}]$, and then a word of cells composes in the reading direction. The same convention makes the endomorphism monoid of a loop polygraph the *opposite* of the monoid its words spell, which is why $\text{Br}(Z) \simeq \mathbf{Braid}^{+, \text{op}}$ rather than $\mathbf{Braid}^+$; word reversal identifies the two. Both $^{\text{op}}$ s are bookkeeping, not content.

2 Background

2.1 Cube chains

We recall two constructions on $\square_*$ from [PZ21; Zie20].

The first is the **wedge** $\vee$. Given $P, Q \in \square_*$, the wedge $P \vee Q$ is the pushout, formed in $\square$, gluing P 's final vertex to Q 's initial vertex:

$$\begin{array}{ccc} I^0 & \xrightarrow{\text{init}_Q} & Q \\ \text{fin}_P \downarrow & \lrcorner & \downarrow \\ P & \longrightarrow & P \vee Q \end{array}$$

Note that the two legs are maps of $\square$, not of $\square_*$. The wedge $P \vee Q$ is bipointed by init_P and fin_Q , and $(\square_*, \vee, I^0)$ is monoidal with unit the point I^0 . A **shape** is a list $a = [a_1, \dots, a_r]$ of *positive* naturals, and its **serial wedge** is

$$\bigvee a := I^{a_1} \vee \dots \vee I^{a_r},$$

with $\bigvee [] = I^0$. Write $\square^\vee \subseteq \square_*$ for the full subcategory of serial wedges. We call the I^{a_i} the **beads** of $\bigvee a$, and put

$$|a| := \sum_i a_i, \quad \deg a := \sum_i (a_i - 1) = |a| - r.$$

The second construction is $\text{Ch}(K)$. For $K \in \square_*$, let $\text{Ch}(K)$ be the full subcategory of the slice $\square_*/K$ spanned by the serial wedges: an object is a bipointed map $c : \bigvee a \rightarrow K$, and a morphism $(\bigvee a \rightarrow K) \rightarrow (\bigvee b \rightarrow K)$ is a bipointed map $\bigvee a \rightarrow \bigvee b$ commuting over K . Such a morphism is called a **refinement**, and it runs from a finer chain to a coarser one. We write $\deg d$ for $\deg a$ where a is the shape of d , and $|d|$ for $|a|$, the number of **events** of d .

Finally, two properties of K from [PZ21] recur. We say K *admits altitude* if there is a function $\text{alt} : K_0 \rightarrow \mathbb{Z}$ with $\text{alt}(\text{fin}(e)) = \text{alt}(\text{init}(e)) + 1$ for every edge $e \in K_1$. Following [PZ21, §2] verbatim, K is *non-self-linked* (NSL) if every $\square$ -map $I^n \rightarrow K$ is injective, for all $n \geq 0$.

We use the following, all from [PZ21].

Lemma 2.1 ([PZ21, §2]). *The $\vee$ operator is a monoidal product on $\square_*$, with unit I^0 .*

Lemma 2.2. $\text{Ch} : \square_* \rightarrow \mathbf{Cat}$ *is a lax monoidal functor from $(\square_*, \vee)$ to $(\mathbf{Cat}, \times)$: the comparison $\text{Ch}(P) \times \text{Ch}(Q) \rightarrow \text{Ch}(P \vee Q)$ concatenates chains. For $u : K \rightarrow L$ the functor $\text{Ch}(u)$ postcomposes the classifying map.*

Lemma 2.3. *When K admits altitude and is NSL, the functor of Lemma 2.2 is an isomorphism of categories, since both categories are skeletal and posets. This is an easy corollary of [PZ21, Lem. 2.11].*

Lemma 2.4. *Serial wedges admit altitude and are NSL. Specifically, the final vertex of $\vee a$ has altitude $|a|$.*

Lemma 2.5 ([PZ21, Prop. 2.7]). *Let P, Q be connected bipointed precubical sets admitting altitude functions. Then every bipointed map $f : P \rightarrow Q$ preserves altitude: $\text{alt}_Q \circ f = \text{alt}_P$.*

2.2 Wedge coordinates

Everything in this paper is bookkeeping about the events of a wedge, so we set that up once.

Lemma 2.6. *A functor $P : \mathbf{Cube} \rightarrow \mathbf{Set}$ extends, by left Kan extension along the Yoneda embedding, to a cocontinuous $E(P) : \square \rightarrow \mathbf{Set}$ with $E(P)(I^n) = P(I^n)$. If $P(I^0) = \emptyset$ then for every serial wedge*

$$E(P)(\vee a) \cong \coprod_i P(I^{a_i}). \quad (1)$$

Proof. Applying $E(P)$ to the pushout defining $A \vee B$ gives $E(P)(A \vee B) = E(P)(A) \sqcup_{P(I^0)} E(P)(B)$, which is the coproduct when $P(I^0) = \emptyset$. $\square$

Definition 2.7. Let $\text{Coord} : \mathbf{Cube} \rightarrow \mathbf{Set}$ send the cube I^n to its set of directions $[n]$, and send a face embedding $g : I^n \rightarrow I^m$ to the monotone injection $g_* : [n] \hookrightarrow [m]$ that names the n directions g leaves free. Since $\text{Coord}(I^0) = [0] = \emptyset$, Lemma 2.6 applies, and we write

$$\text{Ev}(a) := E(\text{Coord})(\vee a) \cong \coprod_i [a_i]$$

for the set of **events** of the shape a . For a morphism $f : \vee a \rightarrow \vee b$ we write $f_* := E(\text{Coord})(f) : \text{Ev}(a) \rightarrow \text{Ev}(b)$.

Definition 2.8. Write $\text{blk}(p)$ for the bead index of an event $p \in \text{Ev}(a)$. By the pushout defining $\vee a$, a morphism $f : \vee a \rightarrow \vee b$ is determined by the maps out of the beads $I^{a_i} \rightarrow \vee b$. But each such image lies in a single I^{b_j} since $a_i > 0$. So it corresponds to an order-preserving face embedding

$$f_i : I^{a_i} \longrightarrow I^{b_{\text{blk}(f)(i)}}.$$

This defines $\text{blk}(f)$, and describes f_* on events:

$$f_*(i, d) = (\text{blk}(f)(i), (f_i)_*(d)), \quad d \in [a_i]. \quad (2)$$

In particular $\text{blk} \circ f_* = \text{blk}(f) \circ \text{blk}$.

Lemma 2.9. *For any morphism $f : \vee a \rightarrow I^m$ in $\square$, the function f_* is injective.*

Proof. Induct on a . The empty case is $f_* : \emptyset \rightarrow [m]$, trivially injective. For the induction case, $f : (\vee a) \vee I^n \rightarrow I^m$, which by the pushout splits into $g : \vee a \rightarrow I^m$ and $h : I^n \rightarrow I^m$ with $f_* = g_* \sqcup h_*$ by Lemma 2.6. Here g_* is injective by the induction hypothesis, and h_* is the monotone injection for the face embedding h .

So it suffices that $\text{im } g_*$ and $\text{im } h_*$ are disjoint. Let $c \in \{0, 1\}^m$ be the image of the glued vertex $\text{fin}(\vee a) = \text{init}(I^n)$. Coordinates only increase along a map into a cube, so every direction g leaves free is already 1 at c , while every direction h leaves free is still 0 at c . $\square$

Lemma 2.10. *For any morphism $f : \bigvee a \rightarrow \bigvee b$ in $\square_*$, the function f_* is bijective and $\text{blk}(f)$ is monotone. Moreover f is determined by f_* .*

Proof. To prove injective and monotone, we go by induction on the length of b , generalising over a . If b is empty, then so is a by Lemma 2.5 and Lemma 2.4. So both $\text{blk}(f)$ and f_* are the empty function.

Now consider $f : \bigvee a \rightarrow I^m \vee \bigvee b$. By Lemma 2.4 and Lemma 2.3, $\text{Ch}(I^m) \times \text{Ch}(\bigvee b) = \text{Ch}(I^m \vee \bigvee b)$. We can view f as an object of $\text{Ch}(I^m \vee \bigvee b)$. So a splits as $a = a_l * a_r$, and f splits as

$$(f_1 : \bigvee a_l \rightarrow I^m, \quad f_2 : \bigvee a_r \rightarrow \bigvee b).$$

Since $f = f_1 \vee f_2$, by Lemma 2.6 we have $f_* = (f_1)_* \sqcup (f_2)_*$. Now $(f_2)_*$ is injective by the induction hypothesis and $(f_1)_*$ is injective by Lemma 2.9; this is a disjoint union of injections with disjoint codomains, hence f_* is injective. For blk , note $\text{blk}(f) = \text{blk}(f_1) \sqcup (1 + \text{blk}(f_2))$, where $\text{blk}(f_1)$ is constantly 1 and $1 + \text{blk}(f_2)$ is monotone by induction and > 1 .

For surjectivity of f_* , we count. By Lemma 2.4 the final vertex of $\bigvee a$ has altitude $|a|$, and likewise for b ; by Lemma 2.5 the bipointed map f preserves altitude so $|a| = |b|$. So f_* is an injective function between finite sets of equal cardinality, hence a bijection.

Finally, f is determined by f_* : by (2), f_* determines $\text{blk}(f)$ and each $(f_i)_*$, and a face embedding is determined by the directions it leaves free. $\square$

Definition 2.11 (strands). Order $\text{Ev}(a)$ lexicographically: bead index first, then position inside the bead. This is a linear order on a set of size $|a|$, and we write

$$\text{str}_a : \text{Ev}(a) \xrightarrow{\sim} [|a|]$$

for the unique order isomorphism. We call $\text{str}_a(p)$ the **strand** of the event p : it is the position of p in the canonical execution that fires the beads in order and, inside each bead, fires the directions in their cube order.

2.3 Polygraphs and presentations

We use 2-polygraphs (Burroni's computads) in the form of [Ara+23], stripped to what is needed.

Definition 2.12. A **2-polygraph** $\mathcal{P}$ consists of a set V of 0-cells, a family $\mathcal{P}_1(x, y)$ of 1-cells for $x, y \in V$, a family $\mathcal{P}_2(x, y)$ of 2-cells, and two *words*

$$\text{src}, \text{tgt} : \mathcal{P}_2(x, y) \longrightarrow \text{paths from } x \text{ to } y$$

in the free category on the 1-cells, written left to right. A morphism of polygraphs is a map in each dimension, commuting with src and tgt on the nose. Write **Poly** for the resulting category.

The **category presented by** $\mathcal{P}$, written $\langle \mathcal{P} \rangle$, is the free category on the 1-cells modulo the congruence generated by $\text{src}(\alpha) = \text{tgt}(\alpha)$ for every 2-cell α ; write $q : \text{Free}(\mathcal{P}_1) \rightarrow \langle \mathcal{P} \rangle$ for the quotient functor. We say $\mathcal{P}$ **presents** a category C when we are given an equivalence $E : \langle \mathcal{P} \rangle \rightarrow C$.

Proposition 2.13 (universal property; [Ara+23, Ch. 2]). *Let C be a category. To give a functor $\langle \mathcal{P} \rangle \rightarrow C$ is to give a map on 0-cells together with an arrow of C for each 1-cell, such that the two words of every 2-cell are sent to the same arrow. Two functors out of $\langle \mathcal{P} \rangle$ agreeing on 0-cells and 1-cells are equal. A morphism of polygraphs which is bijective in every dimension is an isomorphism.*

2.4 Localization

Proposition 2.14 ([GZ67, Ch. I]). *Let C be a category and W a class of morphisms. There is a category $C[W^{-1}]$ and a functor $Q : C \rightarrow C[W^{-1}]$ inverting W , universal: a functor $G : C \rightarrow D$ inverting W factors uniquely as $G = \tilde{G} \circ Q$, and two functors out of $C[W^{-1}]$ agreeing after Q are equal.*

We write frac for the arrows built from cospans: if $m : a \rightarrow e$ lies in W and $f : b \rightarrow e$ is any morphism, then in $C^{\text{op}}[(W^{\text{op}})^{-1}]$ the composite

$$\text{frac}(m, f) := Q(f^{\text{op}}) \circ Q(m^{\text{op}})^{-1} : Q(a) \longrightarrow Q(b)$$

is defined; we read it as “back along the order-preserving leg, forward along the other”. Two such glue: for $l : e_0 \rightarrow e$, $m_0 : b \rightarrow e_0$ in W and $f_0 : c \rightarrow e_0$,

$$\text{frac}(m_0, f_0) \circ \text{frac}(m, l \circ m_0) = \text{frac}(m, l \circ f_0), \quad (3)$$

and $\text{frac}(m, m) = \text{id}$.

2.5 Newman’s lemma, coherent form

For a poset P write $\text{Hasse}(P)$ for its Hasse quiver: vertices the elements, one edge per covering relation $x < y$. A **reading** of P in a category C is a map φ sending each element to an object and each cover to an arrow; it extends to paths by composition.

Theorem 2.15 (Newman’s lemma, coherent form; [Hue80], [Ara+23, Ch. 4]). *Let P be a finite poset and φ a reading of P in C . Suppose that for all $x < y_1$, $x < y_2$ with $y_1 \neq y_2$ and every z with $y_1 \leq z$ and $y_2 \leq z$, there are $j \leq z$ and paths p_i from y_i to j with*

$$\varphi(x < y_1) \varphi(p_1) = \varphi(x < y_2) \varphi(p_2).$$

Then φ sends any two parallel paths of $\text{Hasse}(P)$ to the same arrow, so it extends uniquely to a functor $P \rightarrow C$, and that functor is pinned by its values on covers.

We also need the trivial converse fact about polygons.

Lemma 2.16. *Let P be a finite poset graded by r with $r(y) = r(x) + 1$ for every cover $x < y$, and suppose every element has exactly two neighbours in $\text{Hasse}(P)$ (counting covers up and down). Then two paths out of one element leaving by different covers are the only paths between their endpoints, and if P has a least and a greatest element then $|P| = 2\ell$, where ℓ is the common length of the two maximal chains.*

Proof. Forget the directions: the Hasse graph is 2-regular, hence a disjoint union of cycles. A directed path may not turn back, since r strictly increases along it, so at each vertex it leaves by the edge it did not arrive on: a path is determined by its first edge. If P has a least element $\perp$ and a greatest $\top$, both edges at $\perp$ point up and both at $\top$ point down, so the cycle through $\perp$ meets $\top$ and splits into two directed arcs from $\perp$ to $\top$, of equal length ℓ by the grading; and that cycle is all of P , since every element lies on a path $\perp \leq x \leq \top$. $\square$

3 Squares, runs, and biclosed sets

3.1 Squares and triangles

Definition 3.1. Let $\text{Sq}, \text{Tri} : \mathbf{Cube} \rightarrow \mathbf{Set}$ send I^n to the set of monotone injections $[2] \hookrightarrow [n]$, respectively $[3] \hookrightarrow [n]$; that is, to the pairs and the triples of directions of I^n , with face embeddings acting by their free-direction inclusions. Both vanish at I^0 , so Lemma 2.6 applies, and we write

$$\text{Sq}(a) := E(\text{Sq})(\bigvee a) \cong \prod_i \binom{[a_i]}{2}, \quad \text{Tri}(a) := E(\text{Tri})(\bigvee a) \cong \prod_i \binom{[a_i]}{3}.$$

So a **square** of $\bigvee a$ is an ordered pair of events of one bead: we write $\text{lo}(s), \text{hi}(s) \in \text{Ev}(a)$ for its two events, with $\text{blk}(\text{lo } s) = \text{blk}(\text{hi } s)$ and $\text{str}(\text{lo } s) < \text{str}(\text{hi } s)$. A **triangle** t is a triple of events of one bead; it has three edges, which we write $t_{01}, t_{02}, t_{12} \in \text{Sq}(a)$. Counting,

$$|\text{Sq}(a)| = \sum_i \binom{a_i}{2}. \quad (4)$$

For a chain d of shape a we abbreviate $\text{Sq}(d) := \text{Sq}(a)$, and for a refinement $q : o \rightarrow d$ we write $q_* : \text{Sq}(o) \rightarrow \text{Sq}(d)$ for the induced map, which is injective by Lemma 2.10.

Lemma 3.2. *A square s of $\bigvee b$ lies in the image of q_* , for $q : \bigvee a \rightarrow \bigvee b$, if and only if $q_*^{-1}(\text{lo } s)$ and $q_*^{-1}(\text{hi } s)$ lie in one bead of $\bigvee a$. Moreover, for events u, v of $\bigvee b$ in distinct beads,*

$$\text{blk}(q_*^{-1}u) < \text{blk}(q_*^{-1}v) \iff \text{blk}(u) < \text{blk}(v).$$

Proof. The first claim: if $s = q_*(t)$ then the two events pull back to the two events of t , which share a bead. Conversely if the two pullbacks share a bead i then, since $(q_i)_*$ is a monotone injection, they are the two events of a square t of that bead with $q_*(t) = s$. The second: $\text{blk}(q)$ is monotone (Lemma 2.10), so $\text{blk}(q_*^{-1}u) < \text{blk}(q_*^{-1}v)$ implies $\text{blk}(u) \leq \text{blk}(v)$, hence $\text{blk}(u) < \text{blk}(v)$ as the beads differ; and if $\text{blk}(q_*^{-1}u) \geq \text{blk}(q_*^{-1}v)$ the same argument gives $\text{blk}(v) \leq \text{blk}(u)$. $\square$

3.2 Runs

Definition 3.3. A chain is a **run** when every bead is an edge, equivalently when its degree is 0. Write $\text{Run}(K) \subseteq \text{Ch}(K)$ for the full subcategory of runs.

Lemma 3.4. *$\text{Run}(K)$ is discrete: a refinement between runs is an identity.*

Proof. A refinement $f : a \rightarrow b$ has $|a| = |b|$ (Lemma 2.10), so $\deg b - \deg a$ is the number of beads lost. If both are runs no bead is lost, so $\text{blk}(f)$ is a monotone bijection, hence the identity, and then so is f_* and hence f . $\square$

Definition 3.5 (the firing order). Let $x : \bigvee [1^N] \rightarrow \bigvee a$ be a run over the serial wedge $\bigvee a$ (so $N = |a|$). By Lemma 2.10 it is determined by the bijection $x_* : [N] = \text{Ev}([1^N]) \rightarrow \text{Ev}(a)$, which is monotone on beads. Write

$$\text{step}_x := x_*^{-1} : \text{Ev}(a) \xrightarrow{\sim} [N]$$

and call it the **firing order** of x : the run fires event p at time $\text{step}_x(p)$.

Lemma 3.6. *The firing order of a run of $\vee a$ fires the beads in blocks: if $\text{blk}(p) < \text{blk}(q)$ then $\text{step}_x(p) < \text{step}_x(q)$. Hence a run of $\vee a$ is the same thing as a choice of linear order on $[a_i]$ for each bead i , and the bead map $\text{blk} : (\text{Ev}(a), \text{step}_x) \rightarrow [r]$ is a monotone surjection.*

Proof. $\text{blk} \circ x_* = \text{blk}(x) \circ \text{blk}$ with $\text{blk}(x)$ monotone (Lemma 2.10); since each bead of $[1^N]$ is a singleton, blk on $\text{Ev}([1^N])$ is the identity $[N] \rightarrow [N]$, so $\text{blk}(x) = \text{blk} \circ x_*$ is monotone. Being a monotone surjection onto $[r]$, its fibres are intervals, and they are the beads. The only remaining freedom is the order inside each fibre. $\square$

Definition 3.7. The **reversed squares** of a run x of $\vee a$ are

$$\text{rev}(x) := \{ s \in \text{Sq}(a) : \text{step}_x(\text{hi } s) < \text{step}_x(\text{lo } s) \}.$$

Under Lemma 3.6 this is the inversion set, bead by bead, of x 's orders against the cubes' own.

3.3 Biclosed sets

Definition 3.8. A set $R \subseteq \text{Sq}(a)$ is **biclosed** when for every triangle $t \in \text{Tri}(a)$

$$(t_{01} \in R \text{ and } t_{12} \in R \implies t_{02} \in R) \quad \text{and} \quad (t_{02} \in R \implies t_{01} \in R \text{ or } t_{12} \in R).$$

Equivalently, both R and its complement are closed under the first rule.

Theorem 3.9 (a run is a biclosed set). *$x \mapsto \text{rev}(x)$ is a bijection from the runs of $\vee a$ onto the biclosed subsets of $\text{Sq}(a)$. Reversing every bead's order complements rev , so biclosed sets are closed under complement.*

Proof. By Lemma 3.6 a run is a family of linear orders $<_i$ on $[a_i]$, and a triangle lies in a single bead; so the statement is bead-local. Fix a bead $[m]$ with its cube order $<$. For a linear order $\prec$ on $[m]$, $\text{rev}(\prec) = \{(p, q) : p < q, q \prec p\}$ is its inversion set, and $\prec$ is recovered from it by $p \prec q$ iff $(p < q \text{ and } (p, q) \notin R) \text{ or } (q < p \text{ and } (q, p) \in R)$. So rev is injective; its image is the set of R for which this recipe gives a linear order, and totality is automatic, so the image is cut out by transitivity. Transitivity of $\prec$ fails only on a triple $p < q < r$, and the four cases to check are exactly the two displayed clauses at that triangle. Complementation reverses each $\prec$. $\square$

Remark 3.10. For a single bead of dimension m this is the classical description of inversion sets of permutations of $[m]$, and the inclusion order on them is the right weak order on S_m [BB05, §3.1]. All the order theory below is that statement, bead by bead.

Lemma 3.11. *Order $\text{Sq}(a)$ by $s \prec s'$ when $\text{str}(\text{lo } s) < \text{str}(\text{lo } s')$, or $\text{lo } s = \text{lo } s'$ and $\text{str}(\text{hi } s) < \text{str}(\text{hi } s')$; call this the **standard order**. Every lower set of the standard order is biclosed, and so is every upper set.*

Proof. On a triangle with events $p < q < r$ the edges are $t_{01} = (p, q)$, $t_{02} = (p, r)$, $t_{12} = (q, r)$, and the standard order puts them in the order $t_{01} \prec t_{02} \prec t_{12}$. For a lower set R : if $t_{12} \in R$ then $t_{02} \in R$, which gives the first clause; and $t_{02} \in R$ gives $t_{01} \in R$, the second. For an upper set: $t_{01} \in R$ gives $t_{02} \in R$; and $t_{02} \in R$ gives $t_{12} \in R$. $\square$

4 Crossings, order-preserving refinements, and $\text{Br}(K)$

4.1 Crossings

Definition 4.1. Let $\varphi : \bigvee a \rightarrow \bigvee b$ be a map of serial wedges. The squares φ **crosses** are

$$\text{Cr}(\varphi) := \{ s \in \text{Sq}(b) : \text{blk}(\varphi_*^{-1}(\text{hi } s)) < \text{blk}(\varphi_*^{-1}(\text{lo } s)) \}$$

— the squares of the target whose upper event φ brings from a strictly earlier bead than their lower. For a refinement f of chains we write $\text{Cr}(f)$ for the crossings of its underlying wedge map.

Lemma 4.2 (the cocycle). *For $\varphi : \bigvee a \rightarrow \bigvee b$ and $\psi : \bigvee b \rightarrow \bigvee c$,*

$$\text{Cr}(\psi\varphi) = \psi_*(\text{Cr } \varphi) \sqcup \text{Cr}(\psi),$$

a disjoint union. In particular $|\text{Cr}|$ is additive along composites, and $\text{Cr}(\text{id}) = \emptyset$.

Proof. Take $s \in \text{Sq}(c)$ and split on whether $s \in \text{im } \psi_*$.

If $s = \psi_*(t)$, then $s \notin \text{Cr}(\psi)$: the two events of t share a bead (Lemma 3.2). And $(\psi\varphi)_*^{-1}(\text{lo } s) = \varphi_*^{-1}(\text{lo } t)$ and likewise for hi , so $s \in \text{Cr}(\psi\varphi)$ iff $t \in \text{Cr}(\varphi)$. This also shows $\psi_*(\text{Cr } \varphi)$ and $\text{Cr}(\psi)$ are disjoint.

If $s \notin \text{im } \psi_*$, then $s \notin \psi_*(\text{Cr } \varphi)$, and by Lemma 3.2 the two events $u := \psi_*^{-1}(\text{lo } s)$ and $v := \psi_*^{-1}(\text{hi } s)$ lie in distinct beads of $\bigvee b$; so by the second part of Lemma 3.2 applied to φ , $\text{blk}(\varphi_*^{-1}v) < \text{blk}(\varphi_*^{-1}u)$ iff $\text{blk}(v) < \text{blk}(u)$, i.e. $s \in \text{Cr}(\psi\varphi)$ iff $s \in \text{Cr}(\psi)$. $\square$

4.2 Order-preserving refinements

Definition 4.3. A refinement $f : c \rightarrow d$ in $\text{Ch}(K)$ is **order preserving** when $\text{Cr}(f) = \emptyset$. Write $W = W_K$ for the class of order-preserving refinements.

Proposition 4.4. *For a refinement $f : c \rightarrow d$ the following are equivalent.*

1. $f \in W$.
2. $f_* : \text{Ev}(c) \rightarrow \text{Ev}(d)$ is monotone for the strand orders.
3. $\text{str}_d \circ f_* = \text{str}_c$: every event keeps its position.

Proof. (2) $\Leftrightarrow$ (3): f_* is a bijection between finite chains of equal size (Lemma 2.10), and a monotone bijection of finite chains is the unique isomorphism.

(1) $\Leftrightarrow$ (2): Let $u, v \in \text{Ev}(c)$ with $\text{str}(u) < \text{str}(v)$. If $\text{blk}(u) = \text{blk}(v)$ then f_*u and f_*v lie in one bead of d in the same relative order, as $(f_i)_*$ is a monotone injection, so $\text{str}(f_*u) < \text{str}(f_*v)$. If $\text{blk}(u) < \text{blk}(v)$ then $\text{blk}(f_*u) \leq \text{blk}(f_*v)$, so $\text{str}(f_*u) > \text{str}(f_*v)$ is possible only when the two land in one bead, and then $s := (f_*v, f_*u)$ is a square of d with $\text{blk}(f_*^{-1}\text{hi } s) = \text{blk}(u) < \text{blk}(v) = \text{blk}(f_*^{-1}\text{lo } s)$, that is $s \in \text{Cr}(f)$. Conversely every $s \in \text{Cr}(f)$ arises this way from $u := f_*^{-1}(\text{hi } s)$, $v := f_*^{-1}(\text{lo } s)$. $\square$

Corollary 4.5. *W contains the identities, and $fg \in W$ if and only if $f \in W$ and $g \in W$. Two order-preserving refinements with the same source and target are equal.*

Proof. Closure and two-out-of-three are Lemma 4.2 together with injectivity of ψ_* . For the last claim, Proposition 4.4(3) determines $f_* = \text{str}_d^{-1} \circ \text{str}_c$, and f is determined by f_* (Lemma 2.10). $\square$

Remark 4.6. Concretely, $f \in W$ means f only glues adjacent beads: $\text{blk}(f)$ is a monotone surjection and each f_i is the evident inclusion of the i -th block. The elementary ones are the comparisons

$$\bigvee[c_1, \dots, p, q, \dots, c_r] \longrightarrow \bigvee[c_1, \dots, p+q, \dots, c_r],$$

and W is the multiplicative class they generate. In particular, if c and d have the same event count and every bead boundary of d is one of c , then there is exactly one order-preserving refinement $c \rightarrow d$.

4.3 The braiding

Definition 4.7. The **braiding** of K is

$$\text{Br}(K) := \text{Ch}(K)^{\text{op}}[(W^{\text{op}})^{-1}],$$

the localization of Proposition 2.14, with structure functor $Q : \text{Ch}(K)^{\text{op}} \rightarrow \text{Br}(K)$. For a chain c we write $[c] := Q(c)$. For $u : K \rightarrow L$ the pushforward $\text{Ch}(u)$ carries W_K into W_L (the condition is on the wedge map alone), so it descends to $\text{Br}(u) : \text{Br}(K) \rightarrow \text{Br}(L)$, strictly functorially in u .

5 The runs over a chain

Fix $K \in \square_*$ throughout, and let d be a chain of K of shape a .

Definition 5.1. A **run over d** is a run X of K together with a refinement $X \rightarrow d$. Write $\mathcal{R}(d)$ for the set of them, $\text{run}(x)$ for the run and $\text{ref}(x)$ for the refinement.

Lemma 5.2. *Composing with d 's classifying map is a bijection between the runs of the wedge $\bigvee a$ and $\mathcal{R}(d)$. We transport step and rev along it and write $\text{rev}(x) \subseteq \text{Sq}(d)$, and we order $\mathcal{R}(d)$ by*

$$x \leq y \quad :\Longleftrightarrow \quad \text{rev}(x) \subseteq \text{rev}(y).$$

Proof. A refinement $X \rightarrow d$ over K is a wedge map $\bigvee[1^N] \rightarrow \bigvee a$ such that X 's classifying map is the composite with d 's; so the classifying map is forced and the data is exactly the wedge map, i.e. a run of $\bigvee a$. $\square$

Corollary 5.3. *$\text{rev} : \mathcal{R}(d) \rightarrow \{\text{biclosed subsets of } \text{Sq}(d)\}$ is an isomorphism of posets, where the target is ordered by inclusion. It has a least element $\perp_d$ ($\text{rev} = \emptyset$), a greatest element $\top_d$ ($\text{rev} = \text{Sq}(d)$), and complementation is an anti-automorphism.*

Proof. Theorem 3.9 and Lemma 5.2; $\emptyset$ and $\text{Sq}(d)$ are biclosed, and the complement of a biclosed set is biclosed. $\square$

Definition 5.4. Write $\underline{d} \in \text{Run}(K)$ for the run underlying $\perp_d$ and $\bar{d} \in \text{Run}(K)$ for the run underlying $\top_d$.

Lemma 5.5. $\text{rev}(x) = \text{Cr}(\text{ref}(x))$ for every $x \in \mathcal{R}(d)$.

Proof. Every bead of the source is an edge, so the bead index of an event of the source is its strand, and $\text{blk} \circ (\text{ref}(x))_*^{-1} = \text{step}_x$. Now compare Definition 4.1 with Definition 3.7. $\square$

Definition 5.6. For a refinement $q : o \rightarrow d$, the **push** $q_\# : \mathcal{R}(o) \rightarrow \mathcal{R}(d)$ sends x to $\text{ref}(x)$ followed by q .

Proposition 5.7 (the push formula). *For $q : o \rightarrow d$ and $x \in \mathcal{R}(o)$,*

$$\text{rev}(q_\#x) = q_*(\text{rev}(x)) \sqcup \text{Cr}(q).$$

Consequently $q_^{-1} \text{rev}(q_\#x) = \text{rev}(x)$, so $q_\#$ is an order embedding, and $q_\# \perp_o = \perp_d$ if and only if $q \in W$.*

Proof. This is the cocycle Lemma 4.2 read through Lemma 5.5. For the corollaries: q_* is injective and its image misses $\text{Cr}(q)$ (Lemma 3.2), so $q_*^{-1} \text{rev}(q_\#x) = \text{rev}(x)$ and $q_\#x \leq q_\#y$ iff $x \leq y$; and $\text{rev}(q_\# \perp_o) = \text{Cr}(q)$ is empty exactly when $q \in W$. $\square$

Lemma 5.8. *For $x \in \mathcal{R}(d)$ the following are equivalent: $\text{ref}(x) \in W$; $\text{rev}(x) = \emptyset$; $x = \perp_d$. So $\text{ref}(\perp_d)$ is the only order-preserving refinement into d out of a run, and $\underline{X} = X$ for a run X . Moreover, if $u : c \rightarrow d$ lies in W then $\underline{d} = \underline{c}$.*

Proof. The first equivalences are Lemma 5.5 and Corollary 5.3; the last is Proposition 5.7 applied to u , which gives $u_\# \perp_c = \perp_d$. $\square$

6 Steps, cells, and the paper polygraph

6.1 Steps

Definition 6.1. A square $s \in \text{Sq}(d)$ is a **step** of $x \in \mathcal{R}(d)$ when $\text{rev}(x) \triangle \{s\}$ is biclosed; write $\text{St}(x)$ for the set of steps, and x^s for the **flip**, the run with $\text{rev}(x^s) = \text{rev}(x) \triangle \{s\}$.

Lemma 6.2. *$s \in \text{St}(x)$ if and only if the two events of s are consecutive in the firing order of x .*

Proof. Work in the bead of s ; by Lemma 3.6 its events are consecutive in the firing order, so “consecutive in the bead” and “consecutive overall” agree. If the two events of s are consecutive then toggling s swaps an adjacent pair of a linear order, which is again a linear order, so $\text{rev}(x) \triangle \{s\}$ is biclosed by Theorem 3.9. If some event w of the bead fires strictly between them then in $\text{rev}(x) \triangle \{s\}$ the relative order of $\text{lo } s$ and $\text{hi } s$ is swapped while that of each with w is not, so w would have to lie both before and after the pair: not a linear order. $\square$

Proposition 6.3. $|\text{St}(x)| = \deg d$.

Proof. By Lemma 6.2 the steps are the covers of the firing order, a chain on $|d|$ elements, whose two elements lie in one bead. The bead map is a monotone surjection onto the r beads (Lemma 3.6); a monotone surjection from a finite chain onto a chain of r elements collapses exactly $|d| - r$ covers, so the covers within a bead number $|d| - r = \deg d$. $\square$

6.2 Covers

Lemma 6.4 (the cover lemma). *If $x < z$ in $\mathcal{R}(d)$ then there is $s \in \text{St}(x)$ with $s \notin \text{rev}(x)$ and $s \in \text{rev}(z)$.*

Proof. Write $\prec_x, \prec_z$ for the two firing orders. If z agreed with x on every x -covering pair then $\prec_x \subseteq \prec_z$ by transitivity of $\prec_z$, so $\prec_z = \prec_x$ and $z = x$. So some x -cover $p \prec_x q$ has $q \prec_z p$. Its two events lie in one bead: otherwise $\text{blk}(p) < \text{blk}(q)$, and then every run fires p first (Lemma 3.6), contradicting $q \prec_z p$. So the pair is a square, and a step s of x by Lemma 6.2, with $\{\text{lo } s, \text{hi } s\} = \{p, q\}$. If $\text{lo } s = p$ then $s \notin \text{rev}(x)$ and $s \in \text{rev}(z)$, as wanted. If $\text{lo } s = q$ then $s \in \text{rev}(x) \subseteq \text{rev}(z)$, i.e. $p \prec_z q$, contradicting $q \prec_z p$. $\square$

Corollary 6.5. *$x \leq y$ in $\mathcal{R}(d)$ if and only if $\text{rev}(y) = \text{rev}(x) \cup \{s\}$ for some $s \in \text{St}(x) \setminus \text{rev}(x)$; then $y = x^s$, and $|\text{rev}(y)| = |\text{rev}(x)| + 1$. So $\mathcal{R}(d)$ is graded by $|\text{rev}|$, from 0 at $\perp_d$ to $|\text{Sq}(d)|$ at $\top_d$.*

Proof. If $x < y$, Lemma 6.4 gives such an s , and $x \leq x^s \leq y$ forces $y = x^s$. Conversely such an s gives $x < x^s$ since rev grows by one element and rev is an order isomorphism onto biclosed sets (Corollary 5.3). $\square$

6.3 Factorisations and the cell a run rises into

Lemma 6.6 (factorisations are pinned by their middle). *Let $f : c \rightarrow d$ be a refinement. Two factorisations $c \rightarrow m \rightarrow d$ and $c \rightarrow m' \rightarrow d$ of f with m and m' of the same shape are equal, and both legs agree.*

Proof. A wedge map is determined by its action on events (Lemma 2.10) and is injective there, so it is a monomorphism of $\text{Ch}(K)$; and the first leg $c \rightarrow m$ is determined by f and the shape of m , since blk of it is forced by which events of c fall in which bead of m . $\square$

Proposition 6.7 (a factorisation is a set of steps). *Let $x \in \mathcal{R}(d)$. Sending a factorisation $\text{run}(x) \rightarrow m \rightarrow d$ of $\text{ref}(x)$ to the set of steps of x whose two events lie in one bead of m is a bijection*

$$\text{Fact}(\text{ref}(x)) \xrightarrow{\sim} \{S : S \subseteq \text{St}(x)\}, \quad \deg m = |S|.$$

Write m_S for the middle, $\iota_S : \text{run}(x) \rightarrow m_S$ and $\pi_S : m_S \rightarrow d$ for the legs.

Proof. A middle m makes the $|d|$ events of $\text{run}(x)$ into the beads of m , so it cuts the firing order of x into consecutive intervals; the second leg requires each interval to sit inside one bead of d . Conversely any partition of the firing order into intervals refining the bead partition arises from a chain m whose beads are its blocks, in order, and by Lemma 6.6 the factorisation is pinned by that shape. An interval partition of a finite chain is the set of covers it does not cut, and the covers it does not cut lie inside a bead, so they are steps of x by Lemma 6.2. Finally $\deg m$ is the number of events minus the number of blocks, which is the number of uncut covers, namely $|S|$. $\square$

Definition 6.8 (the cell a run rises into). Say $S \subseteq \text{St}(x)$ **rises** from x when $S \cap \text{rev}(x) = \emptyset$.

Lemma 6.9. *If S rises from x then $\iota_S \in W$; hence $\underline{m}_S = \text{run}(x)$ and m_S is a chain of degree $|S|$ from $\text{run}(x)$ to $\overline{m}_S$.*

Proof. By Lemma 5.5, $\text{Cr}(\iota_S)$ is the set of squares of m_S reversed by x seen over m_S . A square of m_S is a pair of events in one block of the interval partition; the covers inside that block all lie in S , hence are not reversed, so the firing order agrees with the cube order on every consecutive pair of the block, hence — by transitivity of the cube order — on every pair of the block. So $\text{Cr}(\iota_S) = \emptyset$. The rest is Lemma 5.8 and Proposition 6.7. $\square$

Lemma 6.10 (the closure lemma). *Let S rise from x . Then*

$$\text{rev}((\pi_S)_\# \top_{m_S}) = \text{rev}(x) \cup (\pi_S)_*(\text{Sq}(m_S)),$$

and for every $z \geq x$ in $\mathcal{R}(d)$ with $S \subseteq \text{rev}(z)$ we have $(\pi_S)_(\text{Sq}(m_S)) \subseteq \text{rev}(z)$. Hence $(\pi_S)_\# \top_{m_S}$ is the least element of $\mathcal{R}(d)$ above x reversing all of S .*

Proof. The push formula (Proposition 5.7) gives $\text{rev}((\pi_S)_\# \top_{m_S}) = (\pi_S)_* \text{Sq}(m_S) \cup \text{Cr}(\pi_S)$; and the cocycle applied to $\pi_S \circ \iota_S = \text{ref}(x)$ with $\text{Cr}(\iota_S) = \emptyset$ gives $\text{Cr}(\pi_S) = \text{rev}(x)$.

For the second claim, let t be a square of m_S , i.e. a pair of events in one block. Inside a block the firing order of x agrees with the strand order, as in the previous proof, and its covers are exactly S restricted to the block, all of which z reverses. So listing the block as $p_1 < \dots < p_k$ in strand order, z reverses each (p_i, p_{i+1}) ; since $\text{rev}(z)$ is biclosed, the first clause at the triangles (p_i, p_{i+1}, p_{i+2}) , (p_i, p_{i+1}, p_{i+3}) , $\dots$ gives that z reverses every pair of the block, in particular $(\pi_S)_*(t)$. $\square$

Definition 6.11 (cells). For $n \geq 0$ and runs $X, Y \in \text{Run}(K)$, an n -**cell** $X \rightarrow Y$ is a chain e of K with $\deg e = n$, $\underline{e} = X$ and $\bar{e} = Y$. Write $\text{Cell}_n(X, Y)$ for the set of them. So $\coprod_{X, Y} \text{Cell}_n(X, Y)$ is just the set of chains of degree n , sorted by the two runs each one spans; $\text{Cell}_0(X, Y)$ is a point if $X = Y$ and empty otherwise.

Corollary 6.12 (covers are 1-cells). *For a cover $x \leq y$ in $\mathcal{R}(d)$ with step s , the chain $\gamma(x \leq y) := m_{\{s\}}$ is a 1-cell $\text{run}(x) \rightarrow \text{run}(y)$, and $(\pi_{\{s\}})_\# \perp_{m_{\{s\}}} = x$, $(\pi_{\{s\}})_\# \top_{m_{\{s\}}} = y$. Conversely every 1-cell α arises this way from the unique cover $\perp_\alpha \leq \top_\alpha$ of $\mathcal{R}(\alpha)$. So γ , together with $x \mapsto \text{run}(x)$ on vertices, is a map of quivers*

$$\gamma : \text{Hasse}(\mathcal{R}(d)) \longrightarrow (\text{Run}(K), \text{Cell}_1),$$

and it is natural along refinements: $\gamma(q_\# e) = \gamma(e)$ for every refinement $q : o \rightarrow d$ and every cover e of $\mathcal{R}(o)$.

Proof. By Lemma 6.9, $m_{\{s\}}$ has degree 1 and $\underline{m_{\{s\}}} = \text{run}(x)$; by Lemma 6.10 with $S = \{s\}$, $(\pi_{\{s\}})_\# \top_{m_{\{s\}}}$ is the least element above x reversing s , which is $x^s = y$, so $\overline{m_{\{s\}}} = \text{run}(y)$. For the converse, a chain α of degree 1 has $|\text{Sq}(\alpha)| = 1$ by (4), so $\mathcal{R}(\alpha)$ has exactly the two elements $\perp_\alpha \leq \top_\alpha$ and $\text{St}(\perp_\alpha)$ is that one square (Proposition 6.3); the factorisation through it is α itself. For naturality along q : the step of $q_\# e$ is q_* of the step of e (Proposition 5.7), and q_* is a bijection on events carrying the firing order of x to that of $q_\# x$, so the two factorisations cut the same interval partition and have the same middle. $\square$

6.4 The attachment of a 2-cell

Lemma 6.13 (sweeping an order). *Let $\prec$ be a linear order on $\text{Sq}(d)$ all of whose lower sets are biclosed, and let $x_t \in \mathcal{R}(d)$ be the run reversing the first t squares. Then*

$$\perp_d = x_0 \prec x_1 \prec \dots \prec x_{|\text{Sq}(d)|} = \top_d$$

is a maximal chain of $\mathcal{R}(d)$. By Lemma 3.11 the standard order and its dual both qualify.

Proof. Each x_t exists by Corollary 5.3, and $\text{rev}(x_{t+1}) = \text{rev}(x_t) \cup \{s\}$ for the $(t+1)$ -st square s ; two biclosed sets differing by one element are a covering pair for inclusion, so $x_t \prec x_{t+1}$, and $s \in \text{St}(x_t)$ by Corollary 6.5. $\square$

Proposition 6.14 (the polygon). *Let e be a chain with $\deg e = 2$. Then every $x \in \mathcal{R}(e)$ has exactly two steps, hence exactly two neighbours in the Hasse diagram; $\mathcal{R}(e)$ is a polygon with $|\mathcal{R}(e)| = 2|\text{Sq}(e)|$ elements; and the only two maximal chains from $\perp_e$ to $\top_e$ are the sweeps of the standard order and of its dual. Write $\text{src}(e)$ and $\text{tgt}(e)$ for the two words of 1-cells they read off under γ .*

Proof. $|\text{St}(x)| = \deg e = 2$ by Proposition 6.3, and the two flips are the two neighbours (Corollary 6.5, in one direction or the other according to whether the step is reversed). Now apply Lemma 2.16 with the grading $|\text{rev}|$: the two sweeps of Lemma 6.13 leave $\perp_e$ by different covers, since the standard order's least and greatest squares differ, so they are the only two maximal chains and $|\mathcal{R}(e)| = 2|\text{Sq}(e)|$. $\square$

Proposition 6.15 (the two species). *Let $\deg e = 2$. Then either e has one bead of dimension 3, in which case $|\text{Sq}(e)| = 3$ and $\mathcal{R}(e)$ is a hexagon, or e has two beads of dimension 2, in which case $|\text{Sq}(e)| = 2$ and $\mathcal{R}(e)$ is a square. In both cases $\text{src}(e)$ and $\text{tgt}(e)$ have length $|\text{Sq}(e)|$ and differ in their first letter.*

Proof. $\deg e = \sum_i (e_i - 1) = 2$ forces the multiset of bead dimensions above 1 to be either $\{3\}$ or $\{2, 2\}$; then (4) gives $\binom{3}{2} = 3$ and $2\binom{2}{2} = 2$. Lengths and the first letter are Proposition 6.14. $\square$

6.5 The polygraph

Definition 6.16. The **paper polygraph** $\mathcal{P}(K)$ has

- 0-cells $\text{Run}(K)$, a set by Lemma 3.4;
- 1-cells $\mathcal{P}(K)_1(X, Y) := \text{Cell}_1(X, Y)$;
- 2-cells $\mathcal{P}(K)_2(X, Y) := \text{Cell}_2(X, Y)$;
- for a 2-cell e , the two words $\text{src}(e), \text{tgt}(e)$ of Proposition 6.14.

Remark 6.17. The data of $\mathcal{P}(K)$ is nothing but the chains of K of degree 0, 1 and 2: no factorisation, no localization and no choice enters it. The only computation in the definition is the attachment, and by Proposition 6.15 that computation has exactly two answers — a hexagon at a 3-cell of K , a square at two independent 2-cells of K .

Proposition 6.18 (diverging covers close). *The reading γ of Corollary 6.12 satisfies the hypothesis of Theorem 2.15 on $\mathcal{R}(d)$, once 1-cells are read in $\langle \mathcal{P}(K) \rangle$. Hence it extends uniquely to a functor*

$$\Gamma_d : \mathcal{R}(d) \longrightarrow \langle \mathcal{P}(K) \rangle,$$

and $\Gamma_d \circ q_{\sharp} = \Gamma_o$ for every refinement $q : o \rightarrow d$.

Proof. Let $x \leq y_1, x \leq y_2$ be distinct covers with steps $s_1 \neq s_2$, and let z be a common upper bound. The set $S = \{s_1, s_2\}$ rises from x , so by Lemma 6.9 the middle $\alpha := m_S$ is a chain of degree 2, that is a 2-cell, and by Lemma 6.10 $j := (\pi_S)_{\sharp} \top_{\alpha}$ satisfies $j \leq z$.

By Proposition 6.14 the poset $\mathcal{R}(\alpha)$ is a polygon with exactly two maximal chains, which are $\text{src}(\alpha)$ and $\text{tgt}(\alpha)$; they leave $\perp_{\alpha}$ by its two covers. Pushing along π_S carries $\perp_{\alpha}$ to x and $\top_{\alpha}$ to j , carries covers to covers (Proposition 5.7, Corollary 6.5) and keeps the 1-cell each cover reads as (Corollary 6.12). So the two maximal chains of $\mathcal{R}(\alpha)$ push to two paths from x to j leaving by y_1

and by y_2 , whose readings are $\text{src}(\alpha)$ and $\text{tgt}(\alpha)$ — equal in $\langle \mathcal{P}(K) \rangle$, by definition of the presented category.

For naturality, $q_\#$ is monotone (Proposition 5.7), so $\Gamma_d \circ q_\#$ and Γ_o are two functors out of $\mathcal{R}(o)$ agreeing on covers (Corollary 6.12), hence equal by the last clause of Theorem 2.15. $\square$

7 Theorem A: the polygraph presents $\text{Br}(K)$

Theorem A compares two universal properties. A functor out of $\langle \mathcal{P}(K) \rangle$ is a reading of the degree-one chains identifying the two words of every degree-two chain (Proposition 2.13); a functor out of $\text{Br}(K)$ is a functor on $\text{Ch}(K)^{\text{op}}$ inverting the order-preserving refinements (Proposition 2.14). We build each from the other.

7.1 From the cells to the localization

Definition 7.1. A cell $\alpha \in \text{Cell}_n(X, Y)$ carries two legs, namely the refinements of its two extreme runs over itself:

$$\alpha_\perp := \text{ref}(\perp_\alpha) : X \rightarrow \alpha, \quad \alpha_\top := \text{ref}(\top_\alpha) : Y \rightarrow \alpha,$$

the first order preserving (Lemma 5.8). Put

$$E(\alpha) := \text{frac}(\alpha_\perp, \alpha_\top) : [X] \rightarrow [Y] \quad \text{in } \text{Br}(K),$$

and $E(X) := [X]$ on 0-cells.

Lemma 7.2 (telescoping). *Fix a chain d and, for $z \in \mathcal{R}(d)$, put $\text{at}(z) := \text{frac}(\text{ref}(\perp_d), \text{ref}(z)) : [d] \rightarrow [\text{run}(z)]$. Then $\text{at}(\perp_d) = \text{id}$, and for every cover $x \leq y$,*

$$\text{at}(y) = E(\gamma(x \leq y)) \circ \text{at}(x).$$

Hence the reading of any path from $\perp_d$ to z under $E \circ \gamma$ is $\text{at}(z)$.

Proof. $\text{at}(\perp_d) = \text{frac}(m, m) = \text{id}$. For a cover with step s put $\beta := \gamma(x \leq y) = m_{\{s\}}$ and $q := \pi_{\{s\}} : \beta \rightarrow d$. By Corollary 6.12, $x = q_\# \perp_\beta$ and $y = q_\# \top_\beta$, so $\text{ref}(x) = q \circ \beta_\perp$ and $\text{ref}(y) = q \circ \beta_\top$. Now (3) with $l = q$ gives

$$E(\beta) \circ \text{at}(x) = \text{frac}(\beta_\perp, \beta_\top) \circ \text{frac}(\text{ref}(\perp_d), q \circ \beta_\perp) = \text{frac}(\text{ref}(\perp_d), q \circ \beta_\top) = \text{at}(y). \quad \square$$

Proposition 7.3 (soundness). *For every 2-cell e , the two words $\text{src}(e)$ and $\text{tgt}(e)$ have the same image under $E \circ \gamma$. Hence E extends to a functor $E : \langle \mathcal{P}(K) \rangle \rightarrow \text{Br}(K)$ with $E(q(\alpha)) = \text{frac}(\alpha_\perp, \alpha_\top)$ on 1-cells.*

Proof. Both words are paths from $\perp_e$ to $\top_e$ in $\mathcal{R}(e)$ (Proposition 6.14), so both read as $\text{at}(\top_e)$ by Lemma 7.2. Now apply Proposition 2.13. $\square$

7.2 From the chains to the cells

Definition 7.4. Define $\Theta : \text{Ch}(K)^{\text{op}} \rightarrow \langle \mathcal{P}(K) \rangle$ by $\Theta(d) := \underline{d}$ on objects and, for a refinement $u : c \rightarrow d$ of $\text{Ch}(K)$,

$$\Theta(u^{\text{op}}) := \Gamma_d(\perp_d \leq u_\# \perp_c) : \underline{d} \longrightarrow \underline{c},$$

using Proposition 6.18 and the fact that $u_\# \perp_c$ has underlying run $\underline{c}$.

Lemma 7.5. Θ is a functor, and it inverts W^{op} ; indeed $\Theta(u^{\text{op}})$ is an identity whenever $u \in W$.

Proof. Let $v : b \rightarrow c$ and $u : c \rightarrow d$, so that $(u \circ v)^{\text{op}} = v^{\text{op}} \circ u^{\text{op}}$. Since $\perp_c \leq v_{\#} \perp_b$ and $u_{\#}$ is monotone (Proposition 5.7), we get $\perp_d \leq u_{\#} \perp_c \leq u_{\#} v_{\#} \perp_b = (u \circ v)_{\#} \perp_b$, and Γ_d is a functor, so

$$\Theta((u \circ v)^{\text{op}}) = \Gamma_d(u_{\#} \perp_c \leq u_{\#} v_{\#} \perp_b) \circ \Gamma_d(\perp_d \leq u_{\#} \perp_c).$$

The left factor is $\Gamma_c(\perp_c \leq v_{\#} \perp_b) = \Theta(v^{\text{op}})$ by the naturality in Proposition 6.18, and the right factor is $\Theta(u^{\text{op}})$. For $u \in W$ we have $u_{\#} \perp_c = \perp_d$ (Proposition 5.7), so the comparison is an identity. $\square$

Definition 7.6. By Proposition 2.14 and Lemma 7.5, Θ descends uniquely along Q to

$$\Phi : \text{Br}(K) \longrightarrow \langle \mathcal{P}(K) \rangle, \quad \Phi \circ Q = \Theta.$$

7.3 The two readings are inverse

Lemma 7.7. $E \circ \Theta \cong Q$ as functors $\text{Ch}(K)^{\text{op}} \rightarrow \text{Br}(K)$, the component at c being the isomorphism $[c] \cong [c]$ that $\text{ref}(\perp_c) \in W$ becomes in the localization.

Proof. By Lemma 7.2, $E(\Theta(u^{\text{op}})) = \text{at}(u_{\#} \perp_c) = \text{frac}(\text{ref}(\perp_d), u \circ \text{ref}(\perp_c))$. Unfolding frac ,

$$\text{frac}(\text{ref}(\perp_d), u \circ \text{ref}(\perp_c)) = Q((\text{ref}(\perp_c))^{\text{op}}) \circ Q(u^{\text{op}}) \circ Q((\text{ref}(\perp_d))^{\text{op}})^{-1},$$

which is $Q(u^{\text{op}})$ conjugated by the two displayed isomorphisms. $\square$

Lemma 7.8. $\Phi \circ E \cong \text{id}_{\langle \mathcal{P}(K) \rangle}$.

Proof. On a 0-cell, $\Phi(E(X)) = \Theta(X) = \underline{X} = X$ (Lemma 5.8). On a 1-cell $\alpha : X \rightarrow Y$: reading a cospan back, $\Phi(\text{frac}(m, f)) = \Theta(f^{\text{op}}) \circ \Theta(m^{\text{op}})^{-1}$, and $\Theta(m^{\text{op}})$ is an identity for $m \in W$ (Lemma 7.5), so $\Phi(E(\alpha)) = \Theta((\alpha_{\top})^{\text{op}})$. Now $(\alpha_{\top})_{\#} \perp_Y$ is $\alpha_{\top}$ itself, that is $\top_{\alpha}$, so

$$\Theta((\alpha_{\top})^{\text{op}}) = \Gamma_{\alpha}(\perp_{\alpha} \leq \top_{\alpha}) = q(\gamma(\perp_{\alpha} \leq \top_{\alpha})) = q(\alpha)$$

by Corollary 6.12, since $\mathcal{R}(\alpha)$ is the two-element poset $\perp_{\alpha} \leq \top_{\alpha}$. So $\Phi \circ E$ agrees with the identity on 0- and 1-cells, hence is the identity by Proposition 2.13. $\square$

Theorem 7.9 (Theorem A). *For every $K \in \square_*$, with no hypothesis on K , the functor E of Proposition 7.3 is an equivalence*

$$E : \langle \mathcal{P}(K) \rangle \xrightarrow{\cong} \text{Br}(K),$$

with inverse Φ . That is, the runs of K and its chains of degree one and two present $\text{Br}(K)$.

Proof. Lemma 7.8 gives $\Phi E \cong \text{id}$. Composing Lemma 7.7 with Q and using uniqueness of descent (Proposition 2.14) gives $E \Phi \cong \text{id}_{\text{Br}(K)}$. $\square$

Remark 7.10. The comparison is an equivalence and no more: $\text{Br}(K)$ is not skeletal, since a chain and the run below it become isomorphic there but are not equal. The runs are a skeleton of $\text{Br}(K)$.

8 Theorem B: naturality in K

Let $f : K \rightarrow L$ and write $f_\bullet := \text{Ch}(f)$ for the pushforward, which postcomposes the classifying map and changes no shape.

Lemma 8.1. *For every chain e of K there is an order isomorphism $\mathcal{R}(e) \cong \mathcal{R}(f_\bullet e)$ carrying x to the run with the same underlying wedge map, and it preserves rev , hence St , covers, $\perp$, $\top$ and the reading γ . Consequently $\underline{f_\bullet e} = \underline{f_\bullet e}$, $\overline{f_\bullet e} = \overline{f_\bullet e}$, and $f_\bullet$ carries $\text{Cell}_n(X, Y)$ into $\text{Cell}_n(f_\bullet X, f_\bullet Y)$.*

Proof. A run over e is a run of the shape $\vee a$ (Lemma 5.2), and $f_\bullet e$ has the same shape, so the two sets of runs over are the same set of wedge maps; the classifying maps are forced on both sides. Everything in Definition 3.7, Definition 6.1 and Corollary 6.12 is defined from the wedge map alone. $\square$

Theorem 8.2 (Theorem B). $K \mapsto \mathcal{P}(K)$, $f \mapsto \mathcal{P}(f) := f_\bullet$ on cells in every dimension, is a functor $\square_* \rightarrow \mathbf{Poly}$, and for every $f : K \rightarrow L$ the square

$$\begin{array}{ccc} \langle \mathcal{P}(K) \rangle & \xrightarrow{\langle \mathcal{P}(f) \rangle} & \langle \mathcal{P}(L) \rangle \\ E_K \downarrow & & \downarrow E_L \\ \text{Br}(K) & \xrightarrow{\text{Br}(f)} & \text{Br}(L) \end{array}$$

commutes as an equality of functors.

Proof. That $\mathcal{P}(f)$ is a morphism of polygraphs is Lemma 8.1: it sends cells to cells, and because it preserves the order, the steps and γ , it sends the sweep of the standard order on $\text{Sq}(e)$ to the sweep of the standard order on $\text{Sq}(f_\bullet e) = \text{Sq}(e)$, so it preserves src and tgt letter by letter. Functoriality in f is functoriality of Ch .

For the square, both composites agree on 0-cells, both sending X to $[f_\bullet X]$. On a 1-cell α : $\text{Br}(f)$ is the descent of $Q_L \circ f_\bullet^{\text{op}}$, so it reads a cospan upstairs, $\text{Br}(f) \text{frac}(m, g) = \text{frac}(f_\bullet m, f_\bullet g)$. Here $f_\bullet(\alpha_\perp)$ is an order-preserving refinement $f_\bullet X \rightarrow f_\bullet \alpha$, hence equals $(f_\bullet \alpha)^\flat$ by Corollary 4.5, and $f_\bullet(\alpha_\top) = (f_\bullet \alpha)^\sharp$ likewise. So both composites send α to $\text{frac}((f_\bullet \alpha)^\flat, (f_\bullet \alpha)^\sharp)$. By Proposition 2.13 the two functors are equal. $\square$

Remark 8.3. The square commutes strictly, which is more than a presentation is entitled to; the comparison isomorphism of Theorem 1.3 is the identity transformation, and in particular it is the identity at id_K . The reason is Corollary 4.5: an order-preserving refinement is pinned by its ends, so there is no choice to make when transporting a cell's legs.

9 Theorem C: at the point the polygraph is Artin's

9.1 The base

Lemma 9.1. $\text{Ch}(Z) \cong \square^\vee$: an object is a shape and a refinement is a bare map of wedges. A run of Z is its event count; write Z_N for the run on N events. Every cell of Z is a loop: $\underline{e} = \bar{e} = Z|_e$.

Proof. Z is terminal, so every serial wedge has exactly one bipointed map to it and every wedge map commutes over it. A run is a shape all of whose beads are edges, i.e. $[1^N]$. Both $\underline{e}$ and $\bar{e}$ are runs with $|e|$ events (Lemma 2.10). $\square$

Definition 9.2 (cuts). For a shape a with $|a| = N$, put

$$\text{cut}(a) := \{ k \in [N-1] : \text{the events of strand } k \text{ and } k+1 \text{ lie in one bead} \}.$$

For a chain d we write $\text{cut}(d)$ for the cuts of its shape.

Lemma 9.3 ([Sta12, §1.2]). $a \mapsto \text{cut}(a)$ is a bijection from the shapes on N events onto the subsets of $[N-1]$, and $|\text{cut}(a)| = \deg a$. Moreover, for a refinement of shapes on N events, cut only grows, and a refinement $a \rightarrow b$ exists precisely when $\text{cut}(a) \subseteq \text{cut}(b)$.

Proof. A shape on N events is a composition of N ; its bead boundaries are a subset of $[N-1]$, and $\text{cut}(a)$ is the complement. This is the classical correspondence between compositions of N and subsets of $[N-1]$, and the number of cuts is N minus the number of beads, which is $\deg a$. A refinement coarsens the blocking, which deletes boundaries. $\square$

Corollary 9.4. $\text{Cell}_n(Z_N, Z_N) \cong \{ C \subseteq [N-1] : |C| = n \}$, by $e \mapsto \text{cut}(e)$. In particular the 1-cells at Z_N are indexed by $[N-1]$ — write σ_k for the one with $\text{cut} = \{k\}$ — and the 2-cells by the pairs $i < k$ in $[N-1]$.

Definition 9.5. For $i \neq k$ in $[N-1]$ set $\text{cox}(i, k) := 3$ if $|i - k| = 1$ and $\text{cox}(i, k) := 2$ otherwise, the Coxeter exponent of S_N .

Proposition 9.6. Let e be a chain of Z with $\deg e = 2$ and $\text{cut}(e) = \{i < k\}$. Then $|\text{Sq}(e)| = \text{cox}(i, k)$.

Proof. By Proposition 6.15, e has either one bead of dimension 3 — with $|\text{Sq}(e)| = 3$ — or two beads of dimension 2 — with $|\text{Sq}(e)| = 2$. The first happens exactly when three consecutive strands lie in one bead, that is when $k = i + 1$; and $\text{cox}(i, k)$ is 3 in that case and 2 otherwise. $\square$

9.2 Where a run passes a step

Definition 9.7. For $x \in \mathcal{R}(d)$ and $s \in \text{St}(x)$, put $\text{pos}_x(s) := \min(\text{step}_x(\text{lo } s), \text{step}_x(\text{hi } s))$: the run passes the step s between times $\text{pos}_x(s)$ and $\text{pos}_x(s) + 1$ (Lemma 6.2).

Lemma 9.8. Let d be a chain with $|d| = N$. Then

1. $\text{pos}_x : \text{St}(x) \rightarrow \text{cut}(d)$ is a bijection, for every $x \in \mathcal{R}(d)$;
2. flipping keeps a step's position: $\text{pos}_{x^s}(s) = \text{pos}_x(s)$;
3. $\text{pos}_{\perp_d}(s) = \text{str}(\text{lo } s)$.

Proof. (1) The bead map is a monotone surjection whose fibres are consecutive intervals of the firing order of lengths $a_1, \dots, a_r$ (Lemma 3.6); those lengths do not depend on x , so the set of times t with t and $t + 1$ in one bead is $\text{cut}(d)$, whatever x is. A step is a cover of the firing order inside a bead (Lemma 6.2), so pos_x lands in $\text{cut}(d)$ and is injective, distinct covers of a chain having distinct lower ends; and both sides have $\deg d$ elements (Proposition 6.3, Lemma 9.3).

(2) Flipping s exchanges the times of its two events and changes no other time, so the minimum of the two is unchanged.

(3) $\perp_d$ fires in strand order, so $\text{step}_{\perp_d} = \text{str}$ and $\text{str}(\text{lo } s) < \text{str}(\text{hi } s)$. $\square$

Lemma 9.9. *Let d be a chain of Z and $x \leq y$ a cover of $\mathcal{R}(d)$ with step s . Then $\gamma(x \leq y) = \sigma_{\text{pos}_x(s)}$.*

Proof. By Corollary 6.12, $\gamma(x \leq y) = m_{\{s\}}$, whose beads are the blocks of the interval partition of the firing order of x cutting every cover but s (Proposition 6.7). So $m_{\{s\}}$ has one bead of dimension 2, sitting at times $\text{pos}_x(s)$ and $\text{pos}_x(s) + 1$, and all others edges: its cut set is $\{\text{pos}_x(s)\}$. $\square$

9.3 The two words alternate

Proposition 9.10. *Let e be a chain of Z with $\deg e = 2$ and $\text{cut}(e) = \{i < k\}$, and let $\perp_e = x_0 \leq \dots \leq x_{|\text{Sq}(e)|} = \top_e$ be the sweep of the standard order (Lemma 6.13), with s_t the $(t+1)$ -st square. Then*

$$\text{pos}_{x_t}(s_t) = \begin{cases} i & t \text{ even} \\ k & t \text{ odd}, \end{cases}$$

so $\text{src}(e) = \sigma_i \sigma_k \sigma_i \dots$ with $\text{cox}(i, k)$ letters. Dually $\text{tgt}(e) = \sigma_k \sigma_i \sigma_k \dots$ with $\text{cox}(i, k)$ letters.

Proof. By Lemma 9.8(1) each x_t has exactly two steps, passed at i and at k .

Base case. By Lemma 9.8(3), $\text{pos}_{\perp_e}(s) = \text{str}(\text{lo } s)$, and the standard order sorts $\text{Sq}(e)$ by $\text{str}(\text{lo } -)$ first. If e is one bead of dimension 3 on strands $i, i+1, i+2$, its squares in standard order are $(i, i+1), (i, i+2), (i+1, i+2)$, whose lower strands are $i, i, i+1 = k$; if e is two beads of dimension 2 on $\{i, i+1\}$ and $\{k, k+1\}$, they are $(i, i+1), (k, k+1)$ with lower strands i, k . Either way the least square is passed at i and the greatest at k .

Induction. Both s_t and s_{t+1} are steps of x_{t+1} : the first because a flip of a step is a step, the second by Lemma 6.13. By Lemma 9.8(2), $\text{pos}_{x_{t+1}}(s_t) = \text{pos}_{x_t}(s_t)$, and by Lemma 9.8(1) the two steps of x_{t+1} are passed at the two distinct cuts i and k . So $\text{pos}_{x_{t+1}}(s_{t+1})$ is the one of i, k that $\text{pos}_{x_t}(s_t)$ is not.

The word then follows from Lemma 9.9, and its length is $|\text{Sq}(e)| = \text{cox}(i, k)$ by Proposition 9.6. The dual order reverses the standard one, so its least square is the standard order's greatest, passed at k , and the same induction applies. $\square$

9.4 The comparison

Definition 9.11 (Artin's polygraph). Let $\mathcal{P}_{\text{Artin}}$ be the 2-polygraph with one 0-cell for each $N \in \mathbb{N}$; at N , the 1-cells $\sigma_1, \dots, \sigma_{N-1}$ (all loops); and, for each pair $i < k$ in $[N-1]$, one 2-cell with the two words

$$\underbrace{\sigma_i \sigma_k \sigma_i \dots}_{\text{cox}(i, k)} \quad \text{and} \quad \underbrace{\sigma_k \sigma_i \sigma_k \dots}_{\text{cox}(i, k)},$$

that is $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$ when $k = i+1$ and $\sigma_i \sigma_k = \sigma_k \sigma_i$ when $k > i+1$. These are Artin's relations [Art47], so the words at N modulo the 2-cells are the positive braid monoid B_N^+ . Since path composition reverses concatenation of words, $\text{End}(N) = (B_N^+)^{\text{op}}$ and $\langle \mathcal{P}_{\text{Artin}} \rangle = \mathbf{Braid}^{+, \text{op}}$, the category with objects $\mathbb{N}$ and no arrows between distinct objects.

Theorem 9.12 (Theorem C). *There is an isomorphism of polygraphs*

$$\mathcal{P}(Z) \cong \mathcal{P}_{\text{Artin}},$$

cell for cell and word for word.

Proof. On 0-cells, $\text{Run}(Z) \cong \mathbb{N}$ by Lemma 9.1. Every cell of Z is a loop, so both polygraphs are a disjoint union over N of loop polygraphs, and it suffices to compare at each N . On 1-cells, $\text{Cell}_1(Z_N, Z_N) \cong [N - 1]$ by Corollary 9.4, matching $\sigma_1, \dots, \sigma_{N-1}$. On 2-cells, $\text{Cell}_2(Z_N, Z_N)$ is the set of pairs $i < k$ in $[N - 1]$, again by Corollary 9.4, matching Artin’s relations. All three maps are bijections, and Proposition 9.10 says the two words of the 2-cell at $\{i < k\}$ are exactly Artin’s two words for that pair. A morphism of polygraphs bijective in every dimension is an isomorphism (Proposition 2.13). $\square$

Corollary 9.13. $\text{Br}(Z) \simeq \mathbf{Braid}^{+, \text{op}}$. Word reversal is an isomorphism $\mathbf{Braid}^{+, \text{op}} \cong \mathbf{Braid}^+$, so $\text{Br}(Z)$ is the positive braid category.

Proof. Theorem 7.9 at $K = Z$ and Theorem 9.12. $\square$

Remark 9.14. No braid theory is used. The generators are the degree-one chains of Z , the relations are its degree-two chains, and the alternating words come from sweeping $\text{Sq}(e)$ in the two directions. Artin’s presentation is what comes out; in particular the identification of $\mathbf{Braid}^+$ with the Garside germ presentation, which is Matsumoto’s theorem, is not needed here.

Remark 9.15. Combining Theorem 8.2 with Theorem 9.12: the terminal map $!_K : K \rightarrow Z$ induces $\text{Br}(!_K) : \text{Br}(K) \rightarrow \text{Br}(Z) \simeq \mathbf{Braid}^{+, \text{op}}$, a braid-valued invariant of the arrows of $\text{Br}(K)$, computed cell by cell from the polygraph. A degree-one chain of K goes to the generator σ_k where k is the position at which its run passes its one step, and a degree-two chain of K goes to the Artin relation of its two positions. This is the promised sharpening of [PZ21]: the braid is read off the chains of K directly, with no nerve and no realization.

References

- [Ara+23] Dimitri Ara, Albert Burroni, Yves Guiraud, Philippe Malbos, François Métayer, and Samuel Mimram. *Polygraphs: From Rewriting to Higher Categories*. 2023. arXiv: [2312.00429 \[math.CT\]](#).
- [Art47] Emil Artin. “Theory of Braids”. In: *Annals of Mathematics* 48.1 (1947), pp. 101–126. DOI: [10.2307/1969218](#).
- [BB05] Anders Björner and Francesco Brenti. *Combinatorics of Coxeter Groups*. Vol. 231. Graduate Texts in Mathematics. Springer, 2005.
- [GGM15] Stéphane Gaussent, Yves Guiraud, and Philippe Malbos. “Coherent presentations of Artin monoids”. In: *Compositio Mathematica* 151.5 (2015), pp. 957–998. DOI: [10.1112/S0010437X14007842](#).
- [GZ67] Peter Gabriel and Michel Zisman. *Calculus of Fractions and Homotopy Theory*. Vol. 35. Ergebnisse der Mathematik und ihrer Grenzgebiete. Springer, 1967.
- [Hue80] Gérard Huet. “Confluent Reductions: Abstract Properties and Applications to Term Rewriting Systems”. In: *Journal of the ACM* 27.4 (1980), pp. 797–821. DOI: [10.1145/322217.322230](#).
- [PZ21] Jakub Paliga and Krzysztof Ziemiański. *Configuration spaces and directed paths on the final precubical set*. 2021. arXiv: [2103.05336 \[math.AT\]](#).
- [Sta12] Richard P. Stanley. *Enumerative Combinatorics, Volume 1*. 2nd ed. Vol. 49. Cambridge Studies in Advanced Mathematics. Cambridge University Press, 2012.
- [Sto26] Zachary Stone. *Cubical: a Lean 4 formalisation of cube chains and their complexification*. 2026. URL: <https://github.com/zstone1/Cubical>.

- [Zie20] Krzysztof Ziemiański. “Spaces of directed paths on pre-cubical sets II”. In: *Journal of Applied and Computational Topology* 4 (2020), pp. 45–78. DOI: [10.1007/s41468-019-00040-z](https://doi.org/10.1007/s41468-019-00040-z). arXiv: [1901.05206](https://arxiv.org/abs/1901.05206).